

Simulation and Modeling

Lecture 16

Input Data Modeling

Testing the fitted distribution

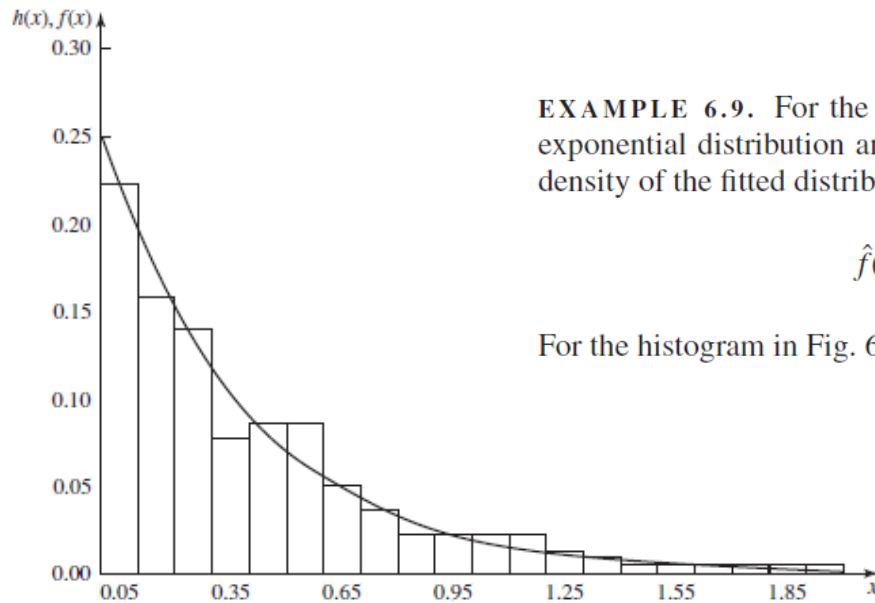
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Testing the fitted distribution

- Heuristic procedure
 - Density-histogram plot
 - Plotting $\nabla \hat{f}(x)$ over the histogram



EXAMPLE 6.9. For the interarrival-time data of Example 6.4, we hypothesized an exponential distribution and obtained the MLE $\hat{\beta} = 0.399$ in Example 6.6. Thus, the density of the fitted distribution is

$$\hat{f}(x) = \begin{cases} 2.506e^{-x/0.399} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

For the histogram in Fig. 6.32(c), we give a density-histogram plot in Fig. 6.35.

FIGURE 6.35

Density-histogram plot for the fitted exponential distribution and the interarrival-time data.

Testing the fitted distribution

- Frequency Comparison
 - Comparison of a histogram with $\hat{f}(x)$
 - ∇b is width of the histogram
 - h_i is the observed proportion of the X_i 's
 - r_j is the expected proportion in the j th interval
 - $r_j = \int_{b_{j-1}}^{b_j} \hat{f}(x) dx$
- Frequency comparison is the plot h_i and r_j

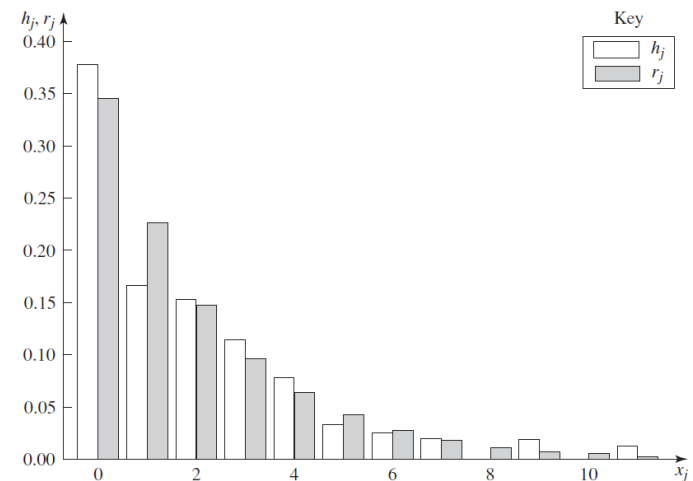


FIGURE 6.36
Frequency comparison for the fitted geometric distribution and the demand-size data.

Testing the fitted distribution

- Distribution-function-difference plot
 - Define a new empirical distribution function $F_n(x)$
 - $F_n(x) = \frac{\text{number of } X_i \leq x}{n}$
 - Proportion of observation that are less than or equal to x
 - Then plot $\hat{F}(x)$ and $F_n(x)$ on the same graph
 - Difference between $\hat{F}(x)$ and $F_n(x)$ are plotted over the range of data

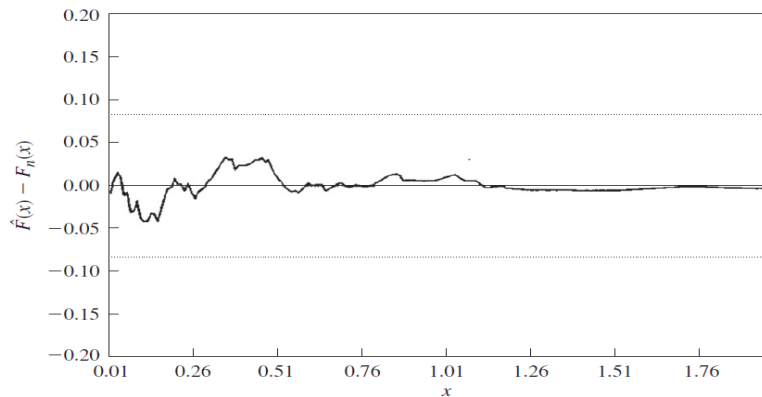


FIGURE 6.37
Distribution-function-differences plot for the fitted exponential distribution and the interarrival-time data.

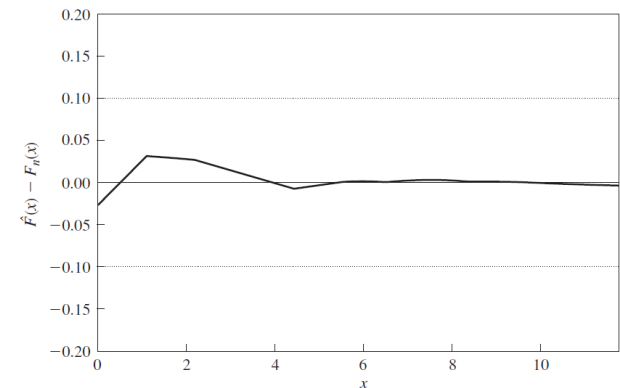


FIGURE 6.38
Distribution-function-differences plot for the fitted geometric distribution and the demand-size data.

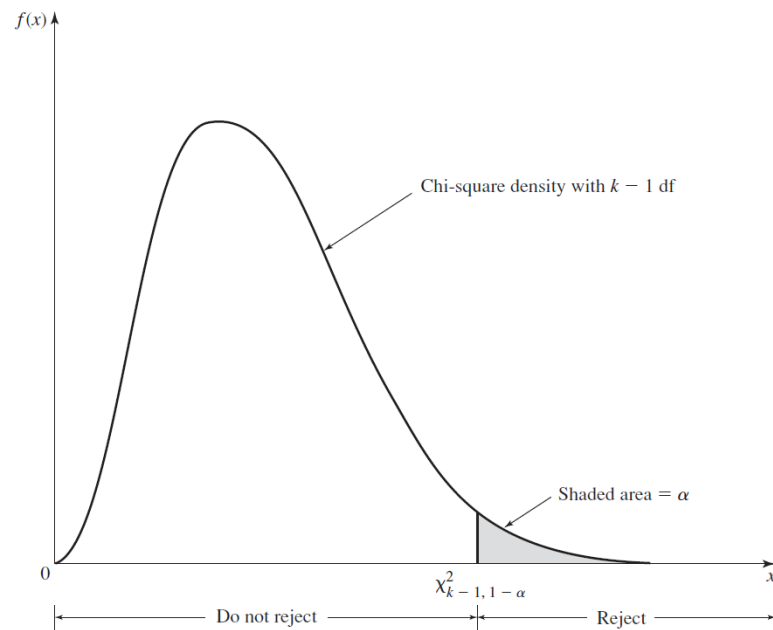
Goodness-of-Fit Tests

- Consider the hypothesis
 - H_0 : The X_i s are IID random variables from $\hat{F}$
- A statistical hypothesis test
 - Determine whether H_0 is acceptable or not
 - Assess $X_1, \dots, X_n$ are independent sample from $\hat{F}$
- Two different tests:
 - Chi-square test
 - Kolmogorov-Smirnov test

Chi-Square Test

- Divide the range of fitted distribution into k adjacent intervals
 - $[a_0, a_1), [a_1, a_2), \dots, [a_{k-1}, a_k)$
 - $N_j = \# \text{ of } X_i \text{ in the } j\text{th interval } [a_j, a_j)$
- Compute the expected proportion of X_i of the j th interval of the fitted distribution $\hat{f}(x)$
 - $p_j = \int_{a_{j-1}}^{a_j} \hat{f}(x) dx$
- Finally, the test statistic is
 - $\chi^2 = \sum_{j=1}^k \frac{(N_j - np_j)^2}{np_j}$
 - H_0 is accepted, if χ^2 is small, otherwise rejected.

distribution.] Then if H_0 is true, χ^2 converges in distribution (as $n \rightarrow \infty$) to a chi-square distribution with $k - 1$ df, which is the same as the gamma $[(k - 1)/2, 2]$ distribution. Thus, for large n , a test with *approximate* level α is obtained by rejecting H_0 if $\chi^2 > \chi_{k-1, 1-\alpha}^2$ (see Fig. 6.45), where $\chi_{k-1, 1-\alpha}^2$ is the upper $1 - \alpha$ critical point for a chi-square distribution with $k - 1$ df. (Values for $\chi_{k-1, 1-\alpha}^2$ can be found in Table T.2 at the end of the book.) Note that the chi-square test is only *valid*, i.e., is of level α , asymptotically as $n \rightarrow \infty$.

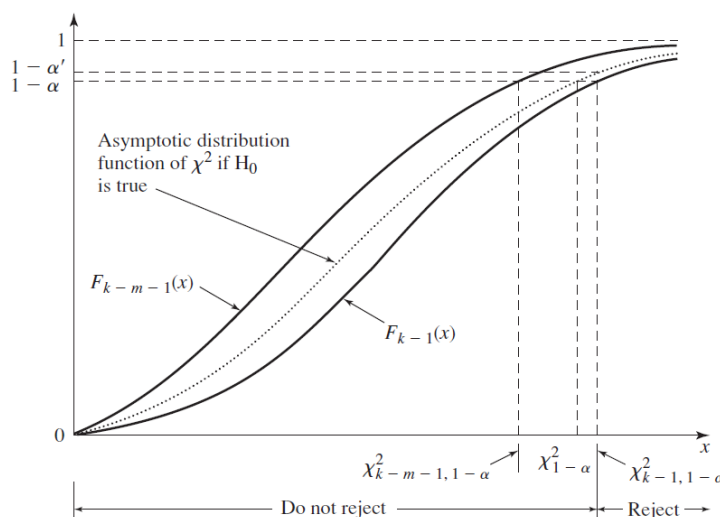


of χ^2 converges to a distribution function that lies *between* the distribution functions of chi-square distributions with $k - 1$ and $k - m - 1$ df. (See Fig. 6.46, where F_{k-1} and F_{k-m-1} represent the distribution functions of chi-square distributions with $k - 1$ and $k - m - 1$ df, respectively, and the dotted distribution function is the one to which the distribution function of χ^2 converges as $n \rightarrow \infty$.) If we let $\chi^2_{1-\alpha}$ be the upper $1 - \alpha$ critical point of the asymptotic distribution of χ^2 , then

$$\chi^2_{k-m-1, 1-\alpha} \leq \chi^2_{1-\alpha} \leq \chi^2_{k-1, 1-\alpha}$$

as shown in Fig. 6.46; unfortunately, the value of $\chi^2_{1-\alpha}$ will not be known in general. It is clear that we should reject H_0 if $\chi^2 > \chi^2_{k-1, 1-\alpha}$ and we should not reject H_0 if $\chi^2 < \chi^2_{k-m-1, 1-\alpha}$; an ambiguous situation occurs when

$$\chi^2_{k-m-1, 1-\alpha} \leq \chi^2 \leq \chi^2_{k-1, 1-\alpha}$$



Kolmogorov-Smirnov Test

- K-S test uses
 - Empirical distribution, $F_n(x) = \frac{\# \text{ of } X_i \leq x}{n}$
 - Fitted distribution, $\hat{F}(x)$
 - Tests the closeness between them
- Test statistic D_n is the largest distance between $F_n(x)$ and $\hat{F}(x)$

$$D_n = \sup_x \{|F_n(x) - \hat{F}(x)|\}$$

- ‘sup’ of a set of numbers A is the smallest value that is greater than or equal to all members of A
- D_n is computed by calculating
 - $D_n^+ = \max_{1 \leq i \leq n} \left\{ \frac{i}{n} - \hat{F}(X_{(i)}) \right\}$ and $D_n^- = \max_{1 \leq i \leq n} \left\{ \hat{F}(X_{(i)}) - \frac{i-1}{n} \right\}$
- Finally, $D_n = \max\{D_n^+, D_n^-\}$

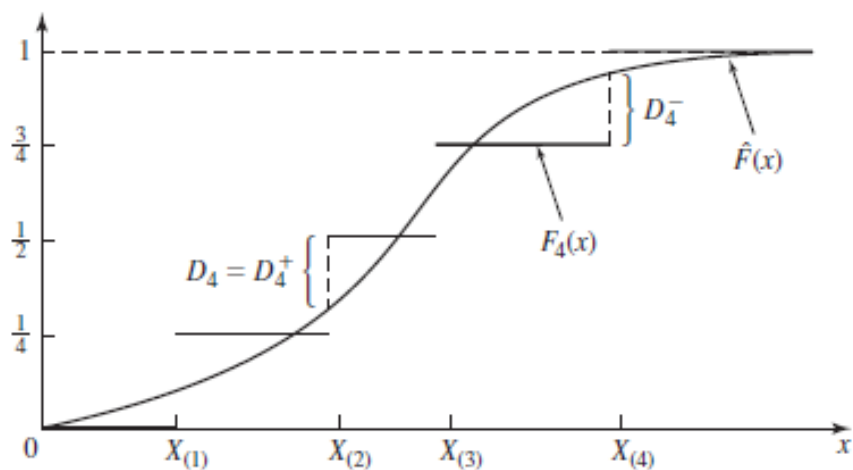


FIGURE 6.47
Geometric meaning of the K-S test statistic D_n for $n = 4$.

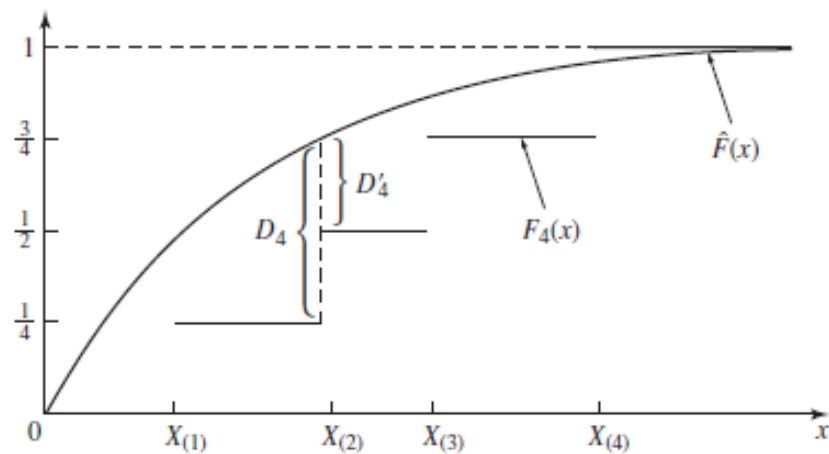


FIGURE 6.48
An example in which the K-S test statistic D_n is not equal to D'_n .

TABLE 6.15

Modified critical values $c_{1-\alpha}$, $c'_{1-\alpha}$, and $c''_{1-\alpha}$ for adjusted K-S test statistics

Case	Adjusted test statistic	$1 - \alpha$				
		0.850	0.900	0.950	0.975	0.990
All parameters known	$\left(\sqrt{n} + 0.12 + \frac{0.11}{\sqrt{n}}\right) D_n$	1.138	1.224	1.358	1.480	1.628
$N(\bar{X}(n), S^2(n))$	$\left(\sqrt{n} - 0.01 + \frac{0.85}{\sqrt{n}}\right) D_n$	0.775	0.819	0.895	0.955	1.035
$\text{expo}(\bar{X}(n))$	$\left(D_n - \frac{0.2}{n}\right) \left(\sqrt{n} + 0.26 + \frac{0.5}{\sqrt{n}}\right)$	0.926	0.990	1.094	1.190	1.308

TABLE 6.16

Modified critical values $c_{1-\alpha}^*$ for the K-S test for the Weibull distribution

n	$1 - \alpha$			
	0.900	0.950	0.975	0.990
10	0.760	0.819	0.880	0.944
20	0.779	0.843	0.907	0.973
50	0.790	0.856	0.922	0.988
∞	0.803	0.874	0.939	1.007

TABLE 6.17

Modified critical values $c_{1-\alpha}^\dagger$ for the K-S test for the log-logistic distribution

n	$1 - \alpha$			
	0.900	0.950	0.975	0.990
10	0.679	0.730	0.774	0.823
20	0.698	0.755	0.800	0.854
50	0.708	0.770	0.817	0.873
∞	0.715	0.780	0.827	0.886

Input data modeling (Step-by-step Process)

Example – Interarrival times

EXAMPLE 6.4. A simulation model was developed for a drive-up banking facility, and data were collected on the arrival pattern for cars. Over a fixed 90-minute interval, 220 cars arrived, and we noted the (continuous) interarrival time X_i (in minutes) between cars i and $i + 1$, for $i = 1, 2, \dots, 219$. Table 6.7 lists these $n = 219$ interarrival times after they have been sorted into increasing order. The numbers of cars arriving in the six consecutive 15-minute intervals were counted and found to be approximately equal, suggesting that the arrival rate is somewhat constant over *this* 90-minute interval. Furthermore, cars arrive one at a time, and there is no reason to believe that the numbers of arrivals in disjoint intervals are not independent. Thus, on theoretical grounds (see Sec. 6.12.1) we postulate that the interarrival times are exponential. To substantiate this hypothesis, we first look at the summary statistics given in Table 6.8. Since $\bar{X}(219) = 0.399 > 0.270 = \hat{x}_{0.5}(219)$ and $\hat{\nu}(219) = 1.478$, this suggests that the underlying distribution is skewed to the right, rather than symmetric. Furthermore, $\hat{c}\hat{\nu}(219) = 0.953$, which is close to the theoretical value of 1 for the exponential distribution. Next we made three different histograms of the data, using $b_0 = 0$ in each case and $\Delta b = 0.050, 0.075$, and 0.100 , as shown in Fig. 6.32 (see also Fig. 6.2). The smoothest-looking

TABLE 6.7
 $n = 219$ interarrival times (minutes) sorted into increasing order

0.01	0.06	0.12	0.23	0.38	0.53	0.88
0.01	0.07	0.12	0.23	0.38	0.53	0.88
0.01	0.07	0.12	0.24	0.38	0.54	0.90
0.01	0.07	0.13	0.25	0.39	0.54	0.93
0.01	0.07	0.13	0.25	0.40	0.55	0.93
0.01	0.07	0.14	0.25	0.40	0.55	0.95
0.01	0.07	0.14	0.25	0.41	0.56	0.97
0.01	0.07	0.14	0.25	0.41	0.57	1.03
0.02	0.07	0.14	0.26	0.43	0.57	1.05
0.02	0.07	0.15	0.26	0.43	0.60	1.05
0.03	0.07	0.15	0.26	0.43	0.61	1.06
0.03	0.08	0.15	0.26	0.44	0.61	1.09
0.03	0.08	0.15	0.26	0.45	0.63	1.10
0.04	0.08	0.15	0.27	0.45	0.63	1.11
0.04	0.08	0.15	0.28	0.46	0.64	1.12
0.04	0.09	0.17	0.28	0.47	0.65	1.17
0.04	0.09	0.18	0.29	0.47	0.65	1.18
0.04	0.10	0.19	0.29	0.47	0.65	1.24
0.04	0.10	0.19	0.30	0.48	0.69	1.24
0.05	0.10	0.19	0.31	0.49	0.69	1.28
0.05	0.10	0.20	0.31	0.49	0.70	1.33
0.05	0.10	0.21	0.32	0.49	0.72	1.38
0.05	0.10	0.21	0.35	0.49	0.72	1.44
0.05	0.10	0.21	0.35	0.50	0.72	1.51
0.05	0.10	0.21	0.35	0.50	0.74	1.72
0.05	0.10	0.21	0.36	0.50	0.75	1.83
0.05	0.11	0.22	0.36	0.51	0.76	1.96
0.05	0.11	0.22	0.36	0.51	0.77	
0.05	0.11	0.22	0.37	0.51	0.79	
0.06	0.11	0.23	0.37	0.52	0.84	
0.06	0.11	0.23	0.38	0.52	0.86	
0.06	0.12	0.23	0.38	0.53	0.87	

TABLE 6.8
Summary statistics for the interarrival-time data

Summary statistic	Value
Minimum	0.010
Maximum	1.960
Mean	0.399
Median	0.270
Variance	0.144
Coefficient of variation	0.953
Skewness	1.478

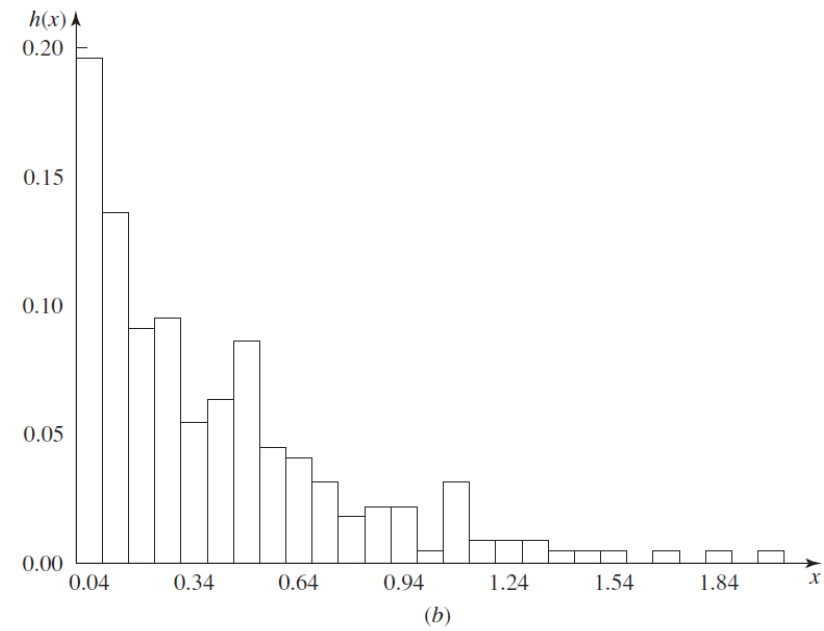
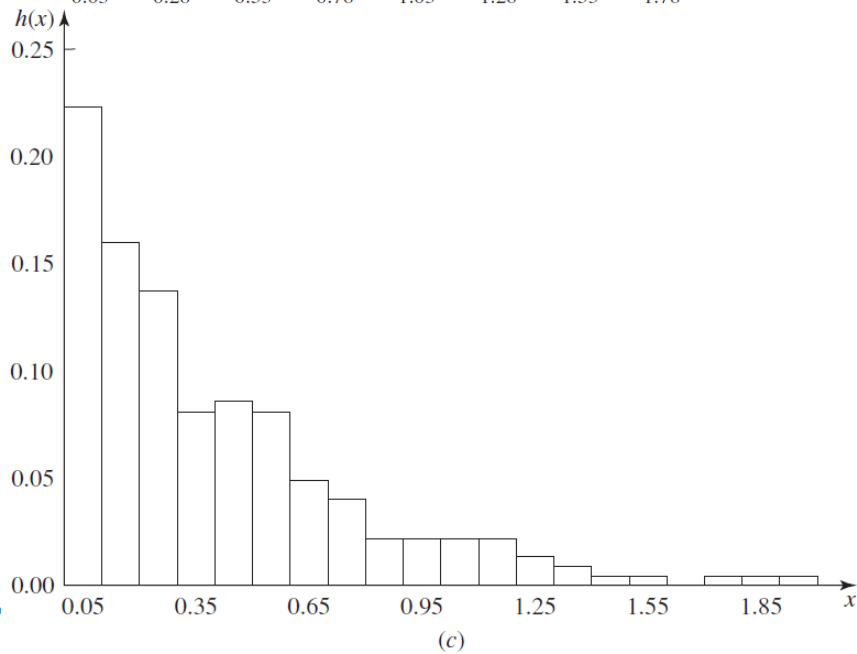
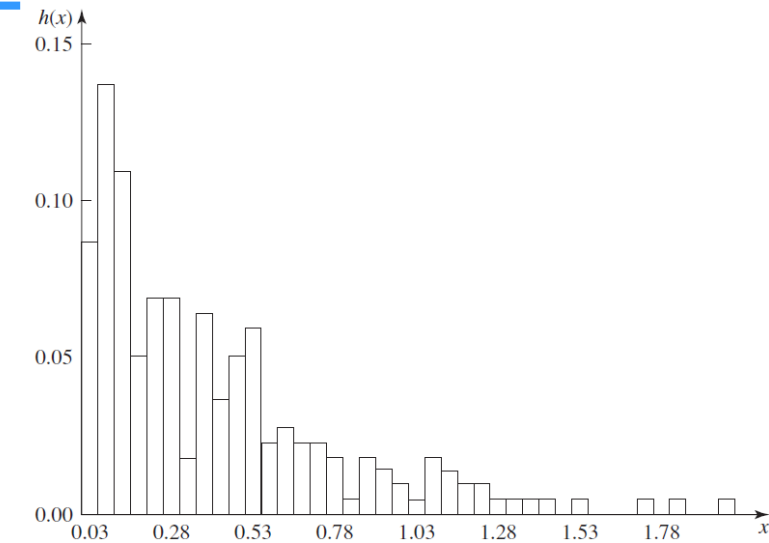


FIGURE 6.32

Histograms of the interarrival-time data in Table 6.7: (a) $\Delta b = 0.050$; (b) $\Delta b = 0.075$; (c) $\Delta b = 0.100$.

Quantile	Depth	Sample value(s)		Midpoint
Median	110	0.270		0.270
Quartiles	55.5	0.100	0.545	0.323
Octiles	28	0.050	0.870	0.460
Extremes	1	0.010	1.960	0.985

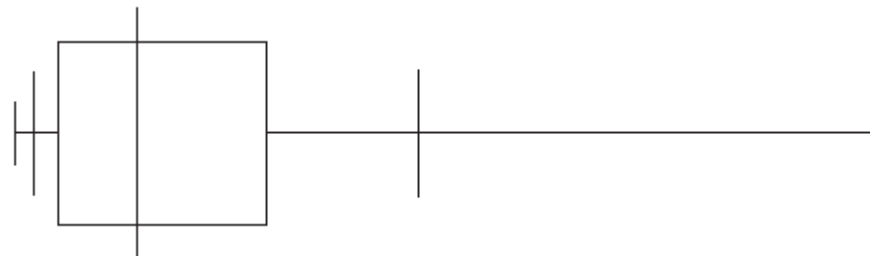


FIGURE 6.33

Quantile summary and box plot for the interarrival-time data.

EXAMPLE 6.6. For the exponential distribution, $\theta = \beta$ ($\beta > 0$) and $f_\beta(x) = (1/\beta)e^{-x/\beta}$ for $x \geq 0$. The likelihood function is

$$\begin{aligned} L(\beta) &= \left(\frac{1}{\beta} e^{-x_1/\beta}\right) \left(\frac{1}{\beta} e^{-x_2/\beta}\right) \cdots \left(\frac{1}{\beta} e^{-x_n/\beta}\right) \\ &= \beta^{-n} \exp\left(-\frac{1}{\beta} \sum_{i=1}^n X_i\right) \end{aligned}$$

and we seek the value of β that maximizes $L(\beta)$ over all $\beta > 0$. This task is more easily accomplished if, instead of working directly with $L(\beta)$, we work with its logarithm. Thus, we define the *log-likelihood function* as

$$l(\beta) = \ln L(\beta) = -n \ln \beta - \frac{1}{\beta} \sum_{i=1}^n X_i$$

Since the logarithm function is strictly increasing, maximizing $L(\beta)$ is equivalent to maximizing $l(\beta)$, which is much easier; that is, $\hat{\beta}$ maximizes $L(\beta)$ if and only if $\hat{\beta}$ maximizes $l(\beta)$. Standard differential calculus can be used to maximize $l(\beta)$ by setting its derivative to zero and solving for β . That is,

$$\frac{dl}{d\beta} = \frac{-n}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^n X_i$$

which equals zero if and only if $\beta = \sum_{i=1}^n X_i / n = \bar{X}(n)$. To make sure that $\beta = \bar{X}(n)$ is a maximizer of $l(\beta)$ (as opposed to a minimizer or an inflection point), a sufficient (but not necessary) condition is that $d^2l/d\beta^2$, evaluated at $\beta = \bar{X}(n)$, be negative. But

$$\frac{d^2l}{d\beta^2} = \frac{n}{\beta^2} - \frac{2}{\beta^3} \sum_{i=1}^n X_i$$

which is easily seen to be negative when $\beta = \bar{X}(n)$ since the X_i 's are positive. Thus, the MLE of β is $\hat{\beta} = \bar{X}(n)$. Notice that the MLE is quite natural here, since β is the mean of the hypothesized distribution and the MLE is the *sample* mean. For the data of Example 6.4, $\hat{\beta} = \bar{X}(219) = 0.399$.

EXAMPLE 6.9. For the interarrival-time data of Example 6.4, we hypothesized an exponential distribution and obtained the MLE $\hat{\beta} = 0.399$ in Example 6.6. Thus, the density of the fitted distribution is

$$\hat{f}(x) = \begin{cases} 2.506e^{-x/0.399} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

For the histogram in Fig. 6.32(c), we give a density-histogram plot in Fig. 6.35.

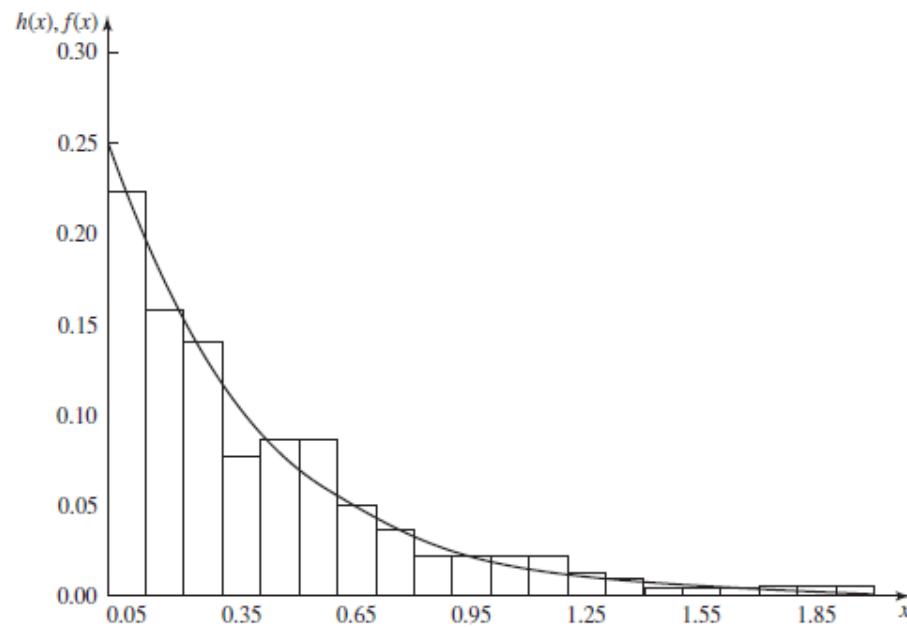


FIGURE 6.35
Density-histogram plot for the fitted exponential distribution and the interarrival-time data.

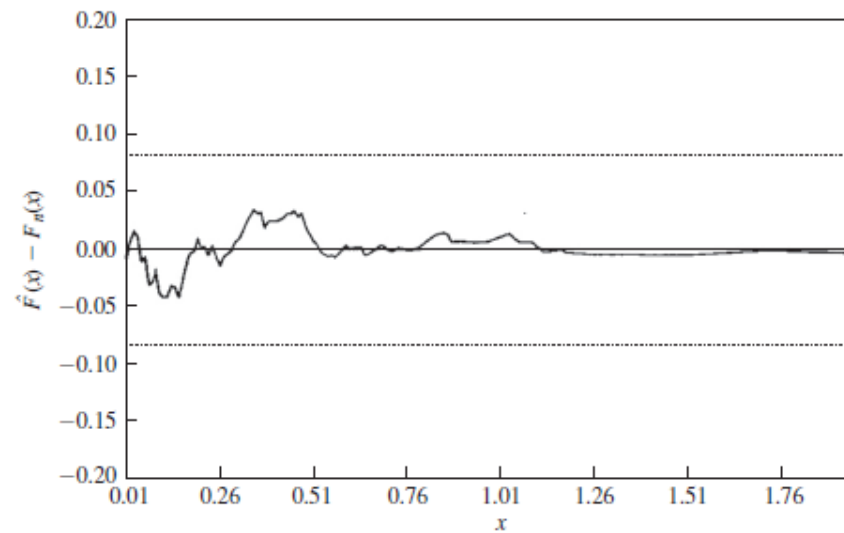


FIGURE 6.37

Distribution-function-differences plot for the fitted exponential distribution and the interarrival-time data.

TABLE 6.12
A chi-square goodness-of-fit test for the interarrival-time data

j	Interval	N_j	np_j	$\frac{(N_j - np_j)^2}{np_j}$
1	[0, 0.020)	8	10.950	0.795
2	[0.020, 0.042)	11	10.950	0.000
3	[0.042, 0.065)	14	10.950	0.850
4	[0.065, 0.089)	14	10.950	0.850
5	[0.089, 0.115)	16	10.950	2.329
6	[0.115, 0.142)	10	10.950	0.082
7	[0.142, 0.172)	7	10.950	1.425
8	[0.172, 0.204)	5	10.950	3.233
9	[0.204, 0.239)	13	10.950	0.384
10	[0.239, 0.277)	12	10.950	0.101
11	[0.277, 0.319)	7	10.950	1.425
12	[0.319, 0.366)	7	10.950	1.425
13	[0.366, 0.419)	12	10.950	0.101
14	[0.419, 0.480)	10	10.950	0.082
15	[0.480, 0.553)	20	10.950	7.480
16	[0.553, 0.642)	9	10.950	0.347
17	[0.642, 0.757)	11	10.950	0.000
18	[0.757, 0.919)	9	10.950	0.347
19	[0.919, 1.195)	14	10.950	0.850
20	[1.195, ∞)	10	10.950	0.082
				$\chi^2 = 22.188$

EXAMPLE 6.15. We now use a chi-square test to compare the $n = 219$ interarrival times of Table 6.7 with the fitted exponential distribution having distribution function $\hat{F}(x) = 1 - e^{-x/0.399}$ for $x \geq 0$. If we form, say, $k = 20$ intervals with $p_j = 1/k = 0.05$ for $j = 1, 2, \dots, 20$, then $np_j = (219)(0.05) = 10.950$, so that this satisfies the guidelines that the intervals be chosen with equal p_j 's and $np_j \geq 5$. In this case, it is easy to find the a_j 's, since $\hat{F}$ can be inverted. That is, we set $a_0 = 0$ and $a_{20} = \infty$, and for $j = 1, 2, \dots, 19$ we want a_j to satisfy $\hat{F}(a_j) = j/20$; this is equivalent to setting $a_j = -0.399 \ln(1 - j/20)$ for $j = 1, 2, \dots, 19$ since $a_j = \hat{F}^{-1}(j/20)$. (For continuous distributions such as the normal, gamma, and beta, the inverse of the distribution function does not have a simple closed form. In these cases, however, F^{-1} can be evaluated by numerical methods; consult the references given in Table 6.11.) The computations for the test are given in Table 6.12, and the value of the test statistic is $\chi^2 = 22.188$. Referring to Table T.2, we see that $\chi^2_{19,0.90} = 27.204$, which is not exceeded by χ^2 , so we would not reject H_0 at the $\alpha = 0.10$ level. (Note that we would also not reject H_0 for certain larger values of α such as 0.25.) Thus, this test gives us no reason to conclude that our data are poorly fitted by the $\text{expo}(0.399)$ distribution.

EXAMPLE 6.18. In Example 6.15 we used a chi-square test to check the goodness of fit of the fitted $\text{expo}(0.399)$ distribution for the interarrival-time data of Table 6.7. We can also apply a K-S test with $\hat{F}(x) = 1 - e^{-x/0.399}$ for $x \geq 0$, by using Case 3 above.

Using the formulas for D_{219}^+ and D_{219}^- , we found that $D_{219} = 0.047$, so that the adjusted test statistic is

$$\left(D_{219} - \frac{0.2}{219}\right)\left(\sqrt{219} + 0.26 + \frac{0.5}{\sqrt{219}}\right) = 0.696$$

Since 0.696 is less than $0.990 = c_{0.90}''$ (from the last row of Table 6.15), we do not reject H_0 at the $\alpha = 0.10$ level.

Example 2 – Demand size

TABLE 6.9

Values and counts for $n = 156$ demand sizes arranged into increasing order

0(59),	1(26),	2(24),	3(18),	4(12),
5(5),	6(4),	7(3),	9(3),	11(2)

TABLE 6.10

Summary statistics for the demand-size data

Summary statistic	Value
Minimum	0.000
Maximum	11.000
Mean	1.891
Median	1.000
Variance	5.285
Lexis ratio	2.795
Skewness	1.687

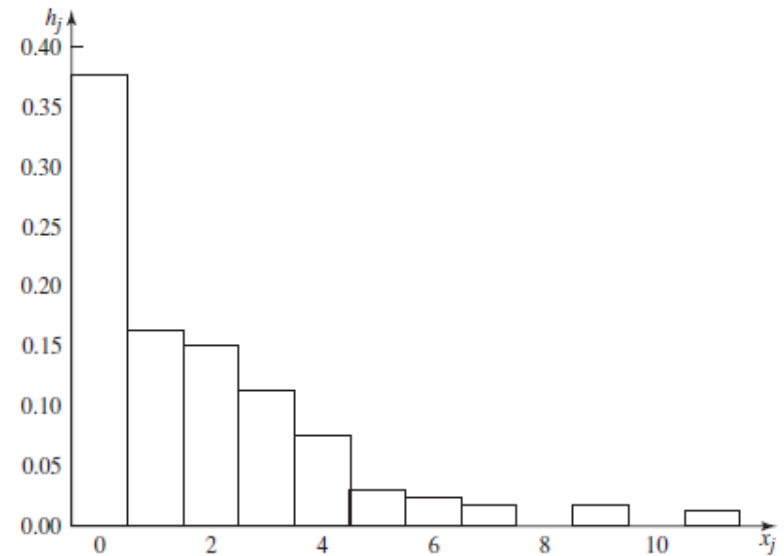


FIGURE 6.34

Histogram of the demand-size data in Table 6.9.

EXAMPLE 6.7. The discrete data of Example 6.5 were hypothesized to come from a geometric distribution. Here $\theta = p$ ($0 < p < 1$) and $p_p(x) = p(1 - p)^x$ for $x = 0, 1, \dots$. The likelihood function is

$$L(p) = p^n (1 - p)^{\sum_{i=1}^n X_i}$$

which is again amenable to the logarithmic transformation to obtain

$$l(p) = \ln L(p) = n \ln p + \sum_{i=1}^n X_i \ln (1 - p)$$

Differentiating $l(p)$, we get

$$\frac{dl}{dp} = \frac{n}{p} - \frac{\sum_{i=1}^n X_i}{1 - p}$$

which equals zero if and only if $p = 1/[\bar{X}(n) + 1]$. To make sure that this is a maximizer, note that

$$\frac{d^2l}{dp^2} = -\frac{n}{p^2} - \frac{\sum_{i=1}^n X_i}{(1 - p)^2} < 0$$

for any valid p . Thus, the MLE of p is $\hat{p} = 1/[\bar{X}(n) + 1]$, which is intuitively appealing (see Prob. 6.9). For the demand-size data of Example 6.5, $\hat{p} = 0.346$.

EXAMPLE 6.10. The demand-size data of Example 6.5 were hypothesized to have come from a geometric distribution, and the MLE of the parameter p was found in Example 6.7 to be $\hat{p} = 0.346$. Thus, the mass function of the fitted distribution is

$$\hat{p}(x) = \begin{cases} 0.346(0.654)^x & \text{if } x = 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

For the histogram in Fig. 6.34, $r_j = \hat{p}(x_j) = \hat{p}(j - 1)$ for $j = 1, 2, \dots, 12$, and the frequency comparison is given in Fig. 6.36, where the h_j 's are represented by the white vertical bars and the r_j 's by the gray vertical bars. Once again, the agreement is good except possibly for $x_2 = 1$.

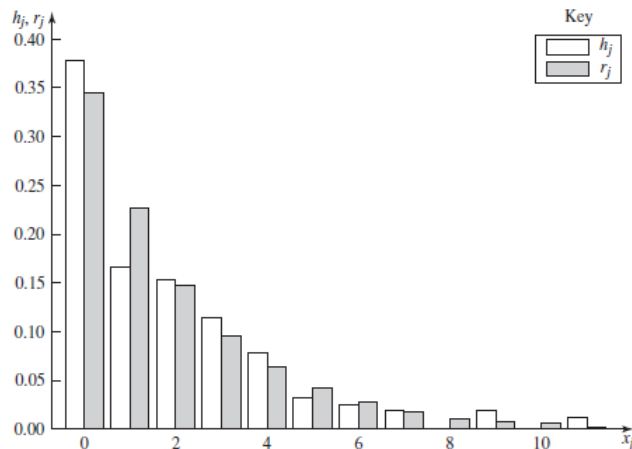


FIGURE 6.36
Frequency comparison for the fitted geometric distribution and the demand-size data.

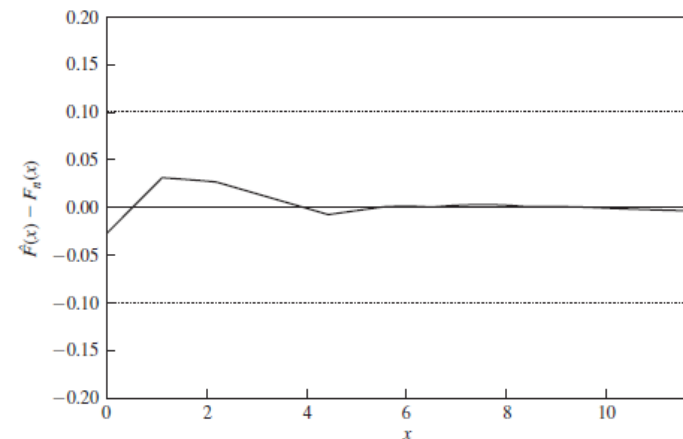


FIGURE 6.38
Distribution-function-differences plot for the fitted geometric distribution and the demand-size data.

EXAMPLE 6.16. As an illustration of the chi-square test in the discrete case, we test how well the fitted $\text{geom}(0.346)$ distribution agrees with the demand-size data of Table 6.9. As is usually the case for discrete distributions, we cannot make the p_j 's exactly equal, but by grouping together adjacent points on which the mass function $\hat{p}(x)$ is defined (here, the nonnegative integers), we can define intervals that make the p_j 's roughly the same. One way to do this is to note that the mode of the fitted distribution is 0; thus, $\hat{p}(0) = 0.346$ is the largest value of the mass function. The large value for the

mode limits our choice of intervals, and we end up with the three intervals given in Table 6.13, where the calculations for the chi-square test are also presented. In particular, $\chi^2 = 1.930$, which is less than the critical value $\chi_{2,0.90}^2 = 4.605$. Thus, we would not reject H_0 at the $\alpha = 0.10$ level, and we have no reason to believe that the demand-size data are not fitted well by a $\text{geom}(0.346)$ distribution.

TABLE 6.13
A chi-square goodness-of-fit test for the demand-size data

j	Interval	N_j	np_j	$\frac{(N_j - np_j)^2}{np_j}$
1	{0}	59	53.960	0.471
2	{1, 2}	50	58.382	1.203
3	{3, 4, ...}	47	43.658	0.256
				$\chi^2 = 1.930$